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Assignment 01: Real-World Software System Analysis

KWSC Unified App (Karachi Water and Sewerage Corporation's Water Services and Tanker Booking App)

Overview:

The KWSC Unified App is the official mobile app of the Karachi Water and Sewerage Corporation (KWSC), launched in 2026 under the idea of "One App, One System, One Karachi". It brings water and sewerage services into one platform where citizens can book water tankers, lodge complaints, view and pay bills, manage site registrations and track the status of their requests. It is available in Urdu, English and Sindhi and uses OTP-based registration. It matters because Karachi's piped water supply falls short of demand, so many households depend on tankers.

A. Purpose:

To modernize KWSC's services by replacing manual, paper- and phone-based processes with a single digital system, so that citizens can request tankers, report problems and pay bills with greater transparency, accountability and real-time visibility.

B. Users:

- Households / residents (book tankers, pay bills, submit complaints)
- Commercial and industrial consumers (register sites and request bulk water)
- KWSC staff and complaint handlers (process requests and complaints)
- Hydrant staff and tanker drivers (receive and fulfil tanker requests)
- KWSC administrators / management (monitor services and performance)

C. Stakeholders:

- Citizens of Karachi (primary stakeholders who depend on water access)
- Karachi Water and Sewerage Corporation (service provider and owner of the app)
- Sindh Government and the Mayor's office (oversight and policy)
- Tanker drivers and hydrant operators (fulfil deliveries)
- Private tanker operators (affected by an official booking channel)
- Banks and payment providers (bill payment)
- App developers and IT support
- Illegal hydrant operators and informal sellers (indirect stakeholders with an interest in resisting transparency)

D. 5 Functional Requirements:

- Users can register and log in using their mobile number with OTP verification.
- Users can book a water tanker by choosing a size, entering their location and submitting a request.
- Users can lodge complaints and support requests and track their status in real time.

- Users can view billing information and pay water and sewerage bills.
- Users can manage site registrations and receive alerts and updates about their services.

E. 5 Non-Functional Requirements:

- Usability: the app must be simple to use and available in Urdu, English and Sindhi for all segments of the population.
- Security and Privacy: OTP verification, and protection of users' personal details such as addresses, phone numbers and payment information.
- Availability: the app should be accessible around the clock, especially during shortages and emergencies.
- Scalability: it must handle very large numbers of users across Karachi, including surges in tanker requests during summer.
- Reliability: request status and alerts must be accurate and consistent with what is actually happening.

F. Risks:

- Booking without supply: a request may be accepted in the app but delayed or unfulfilled because of hydrant shortages, so the app can raise expectations it cannot meet.
- Misuse and corruption: fake or duplicate requests, or favoritism and bribery at hydrants, could bypass the app's fairness and transparency.
- Privacy and data risk: personal data such as addresses and phone numbers could be exposed or shared with third parties.

G. Quality Attributes:

- Transparency (real-time tracking of requests, complaints and bills)
- Accessibility (multilingual interface and simple registration for all citizens)
- Reliability (accurate status updates and dependable service delivery)

H. Where AI Could Be Used:

- Demand forecasting (predicting tanker demand by area and season)
- Smart tanker allocation and route optimization (reducing delivery delays)
- Complaint classification and prioritization (routing complaints to the right team automatically)
- Illegal hydrant and fraud detection (spotting unusual booking or water-loss patterns)
- Multilingual chatbot (helping users book and track in Urdu, English and Sindhi)

I. What Could Go Wrong if AI Is Used Incorrectly:

- Unfair allocation: forecasting based on app usage could favor areas with more smartphone users and neglect poorer neighborhoods.
- Wrong prioritization: a complaint-ranking model could delay urgent problems.
- False fraud flags: genuine citizens or drivers could be wrongly blocked.

- Chatbot errors: wrong answers in Urdu or Sindhi could mislead users about bookings or bills.

Existing Systems and Limitations:

The KWSC Unified App builds on earlier tanker systems. The KWSB Online Tanker System (OTS) connects consumers with hydrant staff, and users are informed of their request by SMS, with the driver's contact and closure of the request. The earlier Water Tanker Management System (WTMS) portal asks for a resident's name, address, CNIC, contact number and tanker size, and delivery is expected within 24 to 48 hours. Private apps such as TankerWala also let users pick a tanker size, drop a location pin, choose a delivery time and rate the experience.

Limitations (analysis):

- ❓ **A booking doesn't guarantee delivery.** The app depends on the physical supply at hydrants. If water or tankers are short, a confirmed request can still be delayed, and the earlier KWSB portal expected delivery in 24 to 48 hours, which is slow for urgent needs.
- ❓ **It covers official KWSC service only.** Much of the tanker water in Karachi is sold by private operators, and the app doesn't appear to control their prices or bring them into one system.
- ❓ **Limited live tracking.** Its public description mentions real-time status tracking of requests and complaints, but not live tanker location or an arrival time like a delivery app would show.
- ❓ **No visible supplier quality checks or ratings.** The public description doesn't mention water-quality verification or customer ratings for tankers.
- ❓ **Digital divide.** Users with low digital literacy, no smartphone or weak internet may keep using informal phone booking. The OTP registration and three languages help, but they don't remove this.
- ❓ **Corruption and misuse risk.** Fake or duplicate requests, or favoritism at hydrants, can still happen outside the app, because the app records requests but can't fully control what happens at the hydrant.
- ❓ **Privacy and data concerns.** Its Play Store listing says it may share some data types with third parties, and users must give personal details such as address and phone number.
- ❓ **Karachi only.** It serves one city, so residents elsewhere in Pakistan can't use it.
- ❓ **Limited service for the poorest.** Free or subsidized tankers may not reach low-income areas through an app-based system.